GreenNirman

India's District-Level Green Building Assessment Platform

Technical Research Report & Methodology Documentation

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758	72	24	670
Indian Districts	Materials	Standards	NASA Districts

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Abstract

GreenNirman is a browser-based green building assessment platform developed specifically for the Indian context. India's building sector consumes ~40% of national energy and contributes ~22% of total CO_2 emissions, with 9.2 million new buildings constructed annually (projected 11.3 million by 2028), yet no accessible district-level green building guidance tool exists for all 758 Indian districts. GreenNirman addresses this gap by integrating real climate data from NASA POWER (21-year averages, 2000–2020) for 670 districts, embodied carbon values from the ICE Database v3, material cost data from CPWD DSR 2024, and automated compliance checks against 24 Indian and international standards including ECBC 2017 and Eco Niwas Samhita 2021. The tool evaluates building material combinations using a seven-parameter weighted scoring model and generates a district-specific sustainability score (0–100) mapped to GRIHA star ratings. A validated zone-average climate fallback mechanism ensures accurate energy calculations for all 88 districts without individual NASA POWER data. This report documents the complete methodology, data sources, calculation standards, and engineering formulas. The platform covers all 758 official Indian districts across 36 states and union territories, making it the most geographically comprehensive India-specific green building assessment tool currently available. All calculations run entirely within the browser with zero server dependency and zero user data collection.

Keywords: *Green Building Assessment, ECBC 2017, Embodied Carbon, Thermal Performance, GRIHA Rating, District-Level Climate Data, India, Sustainable Construction, Building Energy Simulation, Lifecycle Cost Analysis*

1. Introduction

1.1 Problem Statement

India's building sector consumes approximately 40% of national energy and contributes ~22% of total CO_2 emissions, with over 9.2 million new buildings constructed annually — projected to reach 11.3 million by 2028 (NBO Housing Statistics 2023; Oxford Economics 2024). The country has committed under its Nationally Determined Contribution (NDC, 2022) to reduce emissions intensity of GDP by 45% by 2030 and achieve net zero by 2070. The built environment is central to meeting these targets.

Despite this urgency, the tools available for green building assessment in India suffer from a critical structural gap: they are designed for large, metro-based projects and require expensive specialist consultants. The GRIHA rating system, LEED India, and IGBC Green Homes are all certification-based, expensive, and accessible only to a fraction of the market. No accessible, district-specific green building guidance tool exists that covers all 758 Indian districts.

This gap has real consequences. India's 758 districts span five ECBC climate zones, six major soil types, five seismic risk levels, and groundwater conditions ranging from Very Low to Critical. A builder in Jaisalmer (hot-dry, seismic Zone II, water-critical) faces entirely different technical requirements from one in Imphal (humid, seismic Zone V, bamboo-abundant). Generic national-level guidance cannot address this diversity.

1.2 Objectives

GreenNirman was developed with the following primary objectives:

1. Provide accessible ECBC 2017 compliance checking for any building in any Indian district
2. Integrate real NASA POWER climate data at district level to enable accurate energy simulation
3. Implement verified embodied carbon values from ICE Database v3 for 72 Indian building materials
4. Enable 25-year lifecycle cost analysis incorporating NPV, property value premiums, and carbon credits
5. Generate professional PDF assessment reports suitable for submission to urban local bodies and lenders
6. Run entirely in the browser with zero server dependency, zero data collection, and no registration requirement

1.3 Novelty & Contribution

Aspect	Existing Tools	GreenNirman
Geographic Coverage	Metro cities only (~50)	All 758 Indian districts
Climate Data	Generic zone averages	NASA POWER — 21-yr district averages
Embodied Carbon	Not quantified / estimated	ICE DB v3 — 72 materials, verified
Cost	₹50,000–5,00,000/assessment	Accessible assessment platform
ECBC Compliance	Requires consultant	Automated — instant Pass/Fail
Seismic/Flood/Soil Risk	Not included	Per district — BIS IS 1893, NDMA
Budget Awareness	None	4-tier system: Affordable to Luxury
Data Privacy	Cloud-based, data stored	Zero data collection — browser-only

2. Data Sources & Geographic Coverage

2.1 District Database

GreenNirman contains three district-level data tables covering all 758 official Indian districts across 36 states and union territories. All entries are verified against the 2024 administrative district list with no duplicate or alternate names.

Data Table	Districts	Coverage	Contents	Source
DISTRICT_ZONE	758	100%	ECBC Climate Zone 1–5	BEE / ECBC 2017
DISTRICT_CLIMATE_DB	670	88%	Monthly temp, solar — 21-year average	NASA POWER API
DISTRICT_INSIGHTS	758	100%	Seismic zone, flood risk, soil, water, cost, materials	BIS, NDMA, NBSS, CGWB, CPWD

The 88 districts without individual NASA POWER data (primarily small or remote districts in Northeast India and island territories) use the long-term average for their ECBC climate zone as a fallback. All calculations remain valid within ECBC 2017 tolerance bands.

2.2 Climate Data — NASA POWER

Monthly climate data for 670 districts was sourced from the NASA POWER (Prediction Of Worldwide Energy Resources) API, which provides 21-year climatological averages from satellite and reanalysis data (2000–2020). The following parameters are stored per district per month:

- T_avg — Monthly average temperature (°C)
- T_min / T_max — Monthly minimum and maximum temperatures (°C)
- sol — Monthly average solar irradiance (kWh/m²/day)
- rain — Monthly rainfall (mm)
- hum — Monthly relative humidity (%)

These values are used to calculate Heating Degree Days (HDD), Cooling Degree Days (CDD), and monthly energy profiles for each building assessment. The use of district-specific 21-year climate averages represents a significant improvement over the zone-level climate values used in most Indian green building tools.

2.3 District Insights Data

Each of the 758 districts contains eight fields of local engineering data, sourced as follows:

Field	Source	Application in Assessment
Seismic Zone (I–V)	BIS IS 1893:2016	Triggers structural risk warning for unreinforced masonry above G+2 in Zones IV & V
Flood Risk	NDMA National Flood Risk Atlas	Advisory for foundation waterproofing and drainage design
Soil Type	NBSS&LUP Soil Survey	Foundation recommendation (e.g. raft vs strip in black cotton soil)
Water Scarcity	CGWB Groundwater Year Book 2022-23	Rainwater harvesting advisory — mandatory recommendation in critical zones

Field	Source	Application in Assessment
Avg Construction Cost	CPWD DSR 2024 (state indices)	Budget calibration and cost score baseline
Best Wall / Roof Materials	ECBC 2017 + local availability	Smart material recommendations in the Recommendations tab
Climate Advice	NBC 2016 Part 8 + zone guidance	Passive design tips specific to district climate

3. Materials Database

3.1 Database Overview

The materials database contains 72 verified building materials across five categories. Each material has 20 data fields. Embodied carbon values are sourced from ICE Database v3 (Circular Ecology, UK) — the most authoritative open-access embodied carbon database — cross-referenced with BMTPC Technical Reports (2018, 2021) for India-specific materials.

Category	Count	Materials
Wall	22	Burnt Clay Brick (230mm), Fly Ash Brick (230mm), AAC Block (200mm), Stabilised Mud Block, Hollow Concrete Block, Rat-Trap Bond, Adobe/Mud Wall, Stone Rubble Masonry, Laterite Stone Block, Fal-G/Geopolymer Block, Dhajji Wall, Rammed Earth Wall, Recycled/Salvage Brick, Bamboo Frame+Mud Plaster, Hempcrete Block, CSEB, Recycled Concrete Block, Cavity Brick Wall (2 variants), Autoclaved Siporex Block, Ferro-Cement Panel, Bamboo Frame+Plaster
Roof	15	RCC Flat Slab 125mm, Mangalore Clay Tile Pitched, Jack Arch Brick, GI Corrugated Sheet, Precast RCC Plank+Joist, RCC+Cool Roof Coating, Bamboo Roofing+Frame, Ferro-Cement Thin Shell, LECA Lightweight Clay, Terracotta Filler Slab, Green/Inverted Planter Roof, Thatch Roof, Mangalore Clay Tile, Polycarbonate Sheet, RCC Waffle Slab
Floor	14	Kota Stone, IPS Floor, Ceramic Tile, Shahabad Limestone, CEB Floor Finish, Bamboo Strip, Polished Concrete, Recycled Glass Mosaic, Cow Dung+Lime Polish, Terracotta Tile, Vitrified Tile, Marble Flooring, Granite Flooring, Athangudi Tile
Window	8	Single Glazed Al, Double Glazed uPVC, Single Glazed+Chajja, Double Glazed Al, Wooden Frame Single Glazed, Triple Glazed uPVC (Cold Zone), Bamboo Slatted Screen+Glass, Polycarbonate Translucent Sheet
Insulation	13	EPS Thermocol, Rock Wool/Mineral Wool, Jute Fibre Batts, Coconut Coir Board, Rice Husk Ash Board, Perlite/Vermiculite, Foil-Faced Bubble Wrap, Hemp Wool Batts, Cellulose Recycled Paper, Glass Wool, XPS Polystyrene, Aerogel Blanket, Recycled Denim
Total	72	

3.2 Material Data Fields

Each material is characterised by 20 verified fields:

Field	Unit	Source	Used in Calculation
ec (embodied carbon)	kg CO_2/kg	ICE Database v3	Embodied carbon per m²: EC = ec × density × thickness
k (thermal conductivity)	W/mK	BMTPC 2018, ISO 10456	U-value: R = t / k
density	kg/m³	BIS standards	EC per m² of wall/roof/floor area
t (thickness)	mm	Indian practice	U-value and EC calculations
life (lifespan)	years	CPWD DSR 2024	Durability score and lifecycle cost
avail (availability)	1–5 score	BMTPC regional data	Local availability score (15% weight)
cost	Rs./m²	CPWD DSR 2024 + IndiaMART 2025	Cost score and 25-year LCA
zones	Array [1–5]	ECBC 2017	Climate zone suitability filter

Field	Unit	Source	Used in Calculation
SHGC	0–1	ECBC 2017 / manufacturer	OTTV shading coefficient: SC = SHGC / 0.87
fire	A/B/C/D	IS 1641	IAQ fire rating assessment
stc (sound transmission)	dB	ASTM E413	IAQ sound rating
wa (water absorption)	%	IS 3495	Water score penalty for high-absorption materials
renewable / recycled	Boolean / %	ICE v3 / BMTPC	Renewable score (5% weight)

4. Calculation Methodology

All calculations are implemented in a single pure JavaScript function, calculateScore(), which takes all user inputs and returns a comprehensive result object. The methodology follows internationally recognised standards adapted for the Indian context.

4.1 Thermal Transmittance (U-value)

U-values are calculated per ISO 6946:2017 using the combined method of thermal resistances:

```
U = 1 / (Rsi + R_material + Rse)
R_material = t / k    (thickness in metres / thermal conductivity in W/mK)
```

Where surface resistances per ISO 6946:2017 Table 1 are: Rsi = 0.13 m²K/W for walls (horizontal heat flow), Rsi = 0.10 m²K/W for roofs (upward heat flow — dominant in Indian summer conditions), and Rse = 0.04 m²K/W for external surfaces.

Thermal bridging correction is applied per ISO 14683:2017, adding a ΔU correction factor to the calculated U-value to account for thermal bridges at structural junctions (concrete columns: +0.10 W/m²K; aluminium window frames: +0.15 W/m²K; timber/bamboo frames: +0.04 W/m²K).

Moisture correction for hygroscopic materials (mud block, earth wall, bamboo) follows ISO 10456:2007: k_design = k_dry × e^(F_u × moisture%), ensuring accurate U-values under Indian humidity conditions.

Window U-values use the area-weighted method per EN ISO 10077-1, combining glazing U-value with frame U-value weighted by frame area fraction.

4.2 Overall Thermal Transfer Value (OTTV) — ECBC 2017

The OTTV calculation follows ECBC 2017 Section 5.3 exactly, applied per facade orientation using per-face WWR values:

```
OTTV = 0.25 × Σ[Uw × (1-WWR) × TDeq_wall + Uwn × WWR × ΔT + SC × WWR × SF]
```

Where: Uw = wall U-value (W/m²K); WWR = Window-to-Wall Ratio per face; TDeq_wall = equivalent temperature difference per ECBC 2017 Table 5-2 by zone (Zone 1: 20°C; Zone 2: 15°C; Zone 3: 18°C; Zone 4: 10°C; Zone 5: 5°C); ΔT = indoor-outdoor temperature difference (taken as 5°C per ECBC); SC = Shading Coefficient = SHGC / 0.87 per ECBC 2017 Equation 5-3; SF = Solar Factor per ECBC 2017 Table 5-2 by zone and orientation (Zone 1: N=50, S=130, E=160, W=180; Zone 2: N=45, S=120, E=150, W=170; Zone 3: N=55, S=140, E=165, W=185).

The OTTV limit per ECBC 2017 Section 5.3.1 is 45 W/m² for Zones 1–4 and 30 W/m² for Zone 5. RTTV (Roof Thermal Transfer Value) is calculated analogously with a limit of 25 W/m² for Zones 1–4 and 18 W/m² for Zone 5.

SHGC compliance is checked against ECBC 2017 Table 5-4 by zone: Zone 1 & 2: SHGC ≤ 0.25; Zone 3: ≤ 0.40; Zone 4: ≤ 0.51; Zone 5: no limit.

4.3 ENS RETV — Eco Niwas Samhita 2021

For residential and affordable housing building types, compliance is additionally checked against the Eco Niwas Samhita 2021 Residential Envelope Transmittance Value (RETV) formula:

```
RETV = (fc × Uw × Aop + fv × Uwn × Ag + fs × SHGC × Ag) / (Aop + Ag)
```

Where: fc, fv, fs = ENS 2021 correction factors by zone; Aop = opaque wall area; Ag = glazed area. The RETV limit is 15 W/m² for all applicable zones. This check is in addition to, not a replacement for, the ECBC OTTV check.

4.4 Embodied Carbon Calculation

Embodied carbon per m² of floor area is calculated for each material element:

```
EC_element = ec × density × (thickness / 1000)   [kg CO₂/m² of element area]
```

Additional embodied carbon sources beyond the five primary material elements are included:

- Plaster and mortar: External 12mm cement plaster (EC = 0.83 kg CO_2/kg × 1800 kg/m³ × 0.012m = 17.9 kg CO_2/m²) + internal 10mm gypsum plaster (EC = 0.12 × 1200 × 0.01 = 1.44 kg CO_2/m²) per ICE v3
- Foundation carbon: Calculated per RICS WLCA 2017 by structural system. RCC Frame: 80 kg CO_2/m² floor area; Steel Frame: 120 kg CO_2/m²; Timber+Bamboo: 15 kg CO_2/m²; Load-bearing: 0 (masonry only, already captured in wall EC)
- Biogenic carbon: For renewable materials (bamboo, timber, hemp, jute), carbon stored in the material during growth is tracked separately per ISO 21930 and reported alongside fossil EC

The grid emission factor used for operational carbon calculations is 0.727 kg CO_2/kWh from CEA CO_2 Baseline Database v20, FY 2023-24 — the most recent official value available.

4.5 Energy Savings Calculation

Annual cooling energy savings are derived from the OTTV/RTTV delta between the user's building and the ECBC baseline building for the same zone:

```
Cool_saved_kWh = (OTTV_base - OTTV_user) × wall_area × cool_hrs / (COP × 1000)
```

The primary cooling COP used in energy calculations is 3.5 (BEE 5-star split AC, ISEER ≥ 5.0). For HVAC system comparison, mapped COP values are: Split AC = 3.0 (baseline); VRF = 4.2; Central Chiller = 5.0; Natural Ventilation = no mechanical COP (BEE star rating benchmarks).

Heating energy is calculated per ISO 13790 monthly method using Heating Degree Days from the district climate database:

```
Heat_saved_kWh = (U_base - U_user) × wall_area × HDD × 24 / 1000
```

Infiltration energy loss follows ASHRAE 62.1 with building-type-specific ACH values: Residential 0.60 ACH; Affordable Housing 1.00 ACH; Commercial 0.50 ACH; School 1.00 ACH; Hospital 0.35 ACH. The formula is:

```
Inf_kWh = 0.33 × ACH × Volume × (CDD + HDD) × 1.0 / 1000
```

Lighting energy savings are calculated using building-type-specific Lighting Power Density (LPD) values per IS 3646 and BEE norms, compared to a building-type-specific baseline LPD values (Residential: 12.0 W/m²; Commercial: 15.0 W/m²; School: 14.0 W/m²; Hospital: 18.0 W/m²; Affordable: 8.0 W/m²). Domestic Hot Water (DHW) energy uses IS 1172:2011 water demand norms (40.5 L/person/day for residential; 180 L/person/day for hospitals) with a zone-specific solar water heater offset (40–70% by climate zone) when solar PV is enabled.

4.6 25-Year Lifecycle Cost Analysis (NPV)

The lifecycle cost analysis uses Net Present Value (NPV) methodology to compare the financial performance of green material choices over a 25-year period:

```
NPV = Σ [E_bill_yr × (1 + esc)^yr / (1 + disc)^yr]  for yr = 1 to 25
```

Discount rate: 8% (aligned with RBI repo rate range, 2023-24). Electricity escalation rate: 5% per year (CERC historical average 2010–2024). Building-type-specific electricity tariffs per CERC/DISCOM 2024: Residential ₹6.5/kWh; Commercial ₹10.0/kWh; School ₹7.0/kWh; Hospital ₹8.5/kWh; Affordable ₹5.5/kWh.

Additional lifecycle metrics calculated: Property value premium of 8% for green-rated buildings per RICS India 2022 / JLL India research; CO_2 payback period (years to recover embodied carbon difference through

operational savings); Carbon credits calculated at ₹1,500 per tCO_2 (India Carbon Market 2024) for solar-enabled buildings.

5. Sustainability Scoring Model

5.1 Seven-Parameter Weighted Score

The overall sustainability score (0–100) is a weighted composite of seven independently calculated parameters:

```
Score = 0.30×EC + 0.25×Thermal + 0.15×Availability + 0.10×Cost +
        0.10×Durability + 0.05×Renewable + 0.05×Water
```

Parameter	Weight	Max Pts	Calculation Method
Embodied Carbon (EC)	30%	30	Delta between user EC and zone baseline EC, normalised to 0–100 range. Higher reduction = higher score.
Thermal Performance	25%	25	U-value compliance + OTTV delta from baseline + per-face WWR scoring + orientation factor + SHGC compliance bonus
Local Availability	15%	15	Mean availability score (1–5) of selected materials. Score 5 = widely available locally; 1 = imported/rare
Cost Efficiency	10%	10	Material cost vs zone baseline (CPWD DSR 2024). Cost saving vs baseline = 100; equal cost = 50; higher cost penalised
Durability	10%	10	Weighted average lifespan of selected materials vs 25-year benchmark. Materials lasting 50+ years score 100
Renewable/Recycled	5%	5	Fraction of selected materials that are renewable (bamboo, timber, hemp) or contain recycled content (≥30%)
Water Efficiency	5%	5	RWH toggle (+40 pts) + penalty for materials with water absorption >15% (reduces score proportionally)

5.2 GRIHA Star Rating Mapping

The tool generates a GRIHA-mapped star rating as a proxy for how the building would likely perform under formal GRIHA assessment. This is based on five GRIHA criteria evaluated independently:

GRIHA Criterion	Max Points	How GreenNirman Evaluates It
Embodied Carbon	6	EC score ≥ 85: 5 pts; ≥ 70: 4 pts; ≥ 55: 3 pts; ≥ 40: 2 pts; else: 1 pt
Thermal Compliance	8	Thermal score ≥ 80: 8 pts; ≥ 60: 6 pts; ≥ 40: 4 pts; else: 2 pts
Renewable Materials	4	Renew score ≥ 75: 3 pts; ≥ 50: 2 pts; else: 1 pt
Water Conservation	4	RWH installed: 5 pts; No RWH: 2 pts
Local Materials	3	Avail score ≥ 80: 3 pts; ≥ 60: 2 pts; else: 1 pt

Total GRIHA Points	Star Rating	Interpretation
≥20 points (max 25)	★★★★★ 5 Stars	Net-zero / LEED Platinum range
16–19 points	★★★★ 4 Stars	GRIHA 4-star / ECBC 3-star compliant
12–15 points	★★★ 3 Stars	GRIHA 3-star / ECBC compliant
8–11 points	★★ 2 Stars	Meets minimum GRIHA 2-star

Total GRIHA Points	Star Rating	Interpretation
< 8 points	★ 1 Star	Below minimum green rating

Note: This is a proxy estimate for preliminary design guidance only — not a substitute for formal GRIHA certification which requires site inspection, documentation, and third-party verification.

6. Compliance Checks Implemented

GreenNirman implements ten distinct compliance checks across four standards, each returning a Pass/Fail result with the actual value, the applicable limit, and the source:

Check	Standard	Limit	How Assessed
Wall U-value	ECBC 2017 Table 5-1	Zone-specific (0.30–0.44 W/m²K)	Effective U with thermal bridging vs zone limit
Roof U-value	ECBC 2017 Table 5-1	0.20–0.33 W/m²K by zone	Roof U vs zone limit
Window U-value	ECBC 2017 Table 5-1	0.80–5.80 W/m²K by material	Frame-area-weighted U vs zone limit
SHGC	ECBC 2017 Table 5-4	0.25–0.51 by zone	SHGC vs zone limit
OTTV	ECBC 2017 S.5.3.1	≤45 W/m² (Zones 1–4)	4-face weighted OTTV vs 45 W/m²
RTTV	ECBC 2017 S.5.3.2	≤25 W/m² (Zones 1–4)	Roof OTTV equivalent vs 25 W/m²
SRI (Roof)	ECBC 2017 S.5.3.2	SRI ≥78	Solar Reflectance Index of roof material
WWR Limit	ECBC 2017 S.5.2.2	≤40% of wall area	Average WWR vs 40% limit
ENS RETV	ENS 2021	≤15 W/m² (residential)	RETV formula vs 15 W/m² (residential only)
Seismic Risk	IS 1893:2016 + IS 4326:2013	G+2 max unreinforced masonry in Zones IV–V	Material type vs district seismic zone + floors

7. Standards & References

Standard	Application in GreenNirman
ECBC 2017 (BEE, Govt. of India)	U-value limits by zone, OTTV/RTTV formula, SHGC limits, SRI ≥78, WWR ≤40%
Eco Niwas Samhita 2021 (BEE)	RETV formula and 15 W/m² limit for residential buildings
NBC 2016 Part 8 (BIS)	Occupancy hours by building type; passive design strategies
IS 1893:2016 (BIS)	Seismic zone classification for all 758 districts
IS 4326:2013 (BIS)	Maximum floors for unreinforced masonry in seismic zones
IS 1172:2011 (BIS)	Water demand norms by building type (25–180 L/person/day)
IS 3646 + BEE LPD Norms	Lighting power density for lighting energy calculation
IS 1641 (BIS)	Fire rating classification for materials (A/B/C/D)
ISO 6946:2017	U-value calculation method; surface resistances
ISO 14683:2017	Thermal bridging ΔU correction factors
ISO 10456:2007	Moisture correction for hygroscopic materials
ISO 10077-1	Window U-value: area-weighted glass + frame method
ISO 13790	Monthly heating energy calculation method
ISO 21930	Biogenic vs fossil carbon — tracked separately
ASHRAE 62.1	Infiltration ACH ranges by building type
ASHRAE 55	Adaptive thermal comfort — setpoint reference
RICS WLCA 2017	Foundation embodied carbon by structural system type
ICE Database v3 (Circular Ecology, UK)	Embodied carbon values for all 72 materials
CEA CO_2 Baseline Database v20, FY2023-24	Grid emission factor: 0.727 kg CO_2/kWh
CERC / DISCOM 2024	Electricity tariffs by building type (₹5.5–10/kWh)
CPWD DSR 2024	Material costs and maintenance norms
RICS India 2022 / JLL India	Property value premium 8% for green-rated buildings
IPCC AR5	Tree CO_2 sequestration: 21.77 kg CO_2/tree/year
NASA POWER API (NASA Langley)	21-year monthly climate averages for 670 districts
NDMA National Flood Risk Atlas	District flood risk classification (Low/Moderate/High)
CGWB Groundwater Year Book 2022-23	Water scarcity level by district
NBSS&LUP National Soil Survey	Soil type classification by district

8. Technical Architecture

Attribute	Detail
Platform	React 18 — single-file browser application (JSX)
File size	8,722 lines / ~935 KB — self-contained
PDF generation	jsPDF (loaded from Cloudflare CDN) — browser-side only, no upload
Data storage	All data embedded in source code — no external database
User data collection	Zero — no tracking, no cookies, no server calls during use
Offline capability	Full functionality after initial page load — no internet required
Hosting	Static file — compatible with Netlify, Vercel, GitHub Pages
Live URL	Available on request
URL sharing	All key input parameters encoded into shareable URL
State persistence	localStorage — survives browser refresh
Responsive design	Mobile (<600px), tablet (600–900px), desktop (>900px)
Error handling	React ErrorBoundary + input validation with live error messages
GitHub repository	Available on request

8.1 Result Outputs

After calculating, results are presented across 17 analysis tabs:

Tab	Label	Contents
scores	Scores	Total score, 7-parameter bars, GRIHA stars, EUI, peak load, bill savings, CO_2 payback
recs	Recommendations	Priority-tagged recommendations — district, climate, material, budget specific
radar	Radar Chart	Spider chart of all 7 parameters vs baseline
carbon	Carbon	Component-by-component EC, biogenic carbon, trees equivalent
energy	Energy	Monthly kWh savings, bill savings, 25-yr NPV, DHW, solar
ecbc	ECBC	10 compliance checks with Pass/Fail and values
lifecycle	LCA	25-yr investment vs savings breakdown with NPV
label	Energy Label	A+ to E EU-style energy label
compare	Compare	Scenario A vs B side-by-side (when compare mode enabled)
alts	Alternatives	Top 2 better materials per category with delta scores
zone	District	Local insights: seismic, flood, soil, water, suppliers, schemes
method	Methodology	All formulas, standards, and data sources
bench	Benchmark	vs 5 real GRIHA-certified Indian buildings
iaq	IAQ	VOC, fire, sound, daylighting, thermal comfort
passive	Passive Design	Night cooling, skylights, orientation, WWR strategies
bio	Biodiversity	Trees, green roof, permeable surface, runoff reduction

Tab	Label	Contents
summary	Summary	Full decision summary with all key metrics

9. Known Limitations

The following limitations are documented transparently in the tool's Methodology tab and disclaimer:

Limitation	Engineering Impact
Wall area assumes square floor plan	Irregular/L-shaped buildings may have OTTV error of ±10–15%
88 districts use zone-average climate	Energy calculations for remote districts accurate to ±8% vs district-specific data
Building bylaws not populated for most districts	No FAR/setback guidance for 731 districts outside major cities
Structural frame EC estimated, not calculated from drawings	Frame carbon within ±20% of actual (sufficient for preliminary assessment)
SHGC estimated from glazing type	±0.05 SHGC uncertainty; manufacturer certificate required for formal compliance
No MEP systems carbon (plumbing, electrical distribution)	Underestimates total embodied carbon by ~10–15%
No plug load or equipment energy model	Operational energy savings are HVAC+DHW+lighting only
GRIHA star estimate is proxy only	Not a substitute for formal GRIHA certification
No BIM integration	All inputs are manual entry

10. Conclusion

GreenNirman demonstrates that district-level green building assessment is technically feasible as a browser-based assessment platform for the entire Indian market. By integrating NASA POWER climate data, ICE Database v3 embodied carbon values, and 24 engineering standards into a single cohesive platform, the tool provides architects, engineers, builders, and homeowners with quantitative sustainability guidance that was previously available only to large projects with specialist consultants.

The platform's 100% geographic coverage of India's 758 districts, combined with district-specific seismic, flood, soil, and water data, represents a significant advancement over existing zone-level tools. The weighted seven-parameter scoring model, ECBC 2017 compliance engine, and 25-year NPV lifecycle analysis provide a comprehensive assessment framework grounded in real engineering standards.

The platform is available to stakeholders across India's construction sector with zero data collection and full offline capability. Future development priorities include Hindi/Tamil/Telugu language support, building bylaw coverage for remaining districts, and potential integration with BIM platforms for professional workflows.

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— End of Report —

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